

NIU MENGYUAN

(+86) 182 3644 0259 ◇ Jiansgsu Nanjing, CHN

Postcode: 211189 ◇ myniu@seu.edu.cn

OBJECTIVE

Pursue doctoral or research assistant opportunities at the intersection of computer vision and urban planning/geographic information science.

EDUCATION

Master of Information and Communication Engineering, Southeast University Expected 2026

Average Score:90.2/100, GPA:3.6/4

Thesis: Spatial Intelligence-Based Generative Design of 3D Urban Building Morphology

Bachelor of Communication Engineering, Zhengzhou University 2018 - 2022

Average Score:84.64/100, GPA:3.35/4(Rank:9/61, Top 20%)

RESEARCH EXPERIENCE

Automated Urban Scene Generation Jun 2025 - present

- Proposed an enhanced graph representation for modeling urban building layouts, fusing the geometric and semantic attributes of block to enable conditional generation of diverse 2.5D vector urban models.
- Implemented a two-stage block generation method, leveraging LLM and text/image-to-3D to plan scene-level bounding box layouts and the generation of photorealistic building models.
- Employed DiT to generate paired urban semantic and height maps, enabling the precise division of functional zones and facilitating unconditional urban planning.

Procedural Urban Design and Built Environment Optimization Dec 2023 - present

- Designed the intelligent building generation algorithms for various functional zones and integrated front-end and back-end systems to develop a data visualization platform for a three-stage parametric urban design pipeline.
- Utilized LLM enhanced by GraphRAG for knowledge-based Q&A, systematically inducing both rigid and flexible rules of architectural arrangement from extensive databases and real-world urban scenarios.

Spatial Gene Construction within Taihu Lake Basin Apr 2024 - Mar 2025

- Proposed a spatial graph to represent housing land and their surroundings(water, forest, farmland), performing classification based on subgraph similarity and aggregated global graph features through GVAE.
- Extracted road and water system textures from satellite imagery using a fine-tuned SAM model. Performed regional-level statistical analysis on the extracted features to reveal underlying spatial patterns.

Spatial Genealogy throughout China Dec 2024 - Sep 2025

- Established a comprehensive static spatial genealogy throughout China by applying statistical methods. Applied spatial regression and machine learning models to regress township vitality against key indicators.
- Expanded upon the static spatial genealogy to construct a spatiotemporal framework, observing and analyzing the dynamic evolution.

PUBLICATION

1. **Mengyuan Niu**, Xinxin Zhuo, et.al., “Controllable Generation of Large-Scale 3D Urban Layouts with Semantic and Structural Guidance.” SIGSPATIAL, Urban Ai Workshop(Accepted)
2. Chunliang Hua, Daijun Chen, **Mengyuan Niu**, et.al., “Contrastive Learning with Image Deformation and Refined NT-Xent Loss for Urban Morphology Discovery.” ISPRS International Journal of Geo-Information. 2025; 14(5):196. (published)

3. Xinxin Zhuo, **Mengyuan Niu**, Ruizhe Wang, et.al., “Quantifying the Economic Vitality of Urban Streets through Multimodal Data Fusion.” (Under Review)
4. Ruizhe Wang, Chunliang Hua, Ying Shi, **Mengyuan Niu**, et.al., “Enhancement of 3D Gaussian Splatting using Raw Mesh for Photorealistic Recreation of Architectures.” ICPRS(Accepted)

PATENT

Mengyuan Niu, Man Feng, Ruizhe Wang, Xinxin Zhuo, Method and System for Text-Based Smart Scene Generation of City Blocks. (2025117799946)

Ruizhe Wang, Chunliang Hua, **Mengyuan Niu**, A Model Appearance Synthesis Method Based on 3D Gaussians and Shell Mapping. (CN11939026A)

SKILLS

- **Languages:** Python, C++, C, Java, Matlab, Javascript, IELTS 7(6)
- **Libraries:** PyTorch, OpenCV, Geopandas
- **Tools & OS:** Linux, Git, Office, Arcgis, Qgis, Open Street Map, Blender, Google Earth Engine